

Chapter 4: Correlation and Regression

A Level Statistics Notes · Rexponential

Why this chapter matters

So far, each chapter has looked at one variable at a time. This chapter looks at the relationship between two variables: whether they tend to move together, how strongly, and how to model that relationship with a line so that one variable can be used to predict the other. This includes fitting a straight line directly, and also transforming a non-linear relationship into a linear one first — a technique that reappears later in the course whenever data doesn't naturally fit a straight line.

4.1 Scatter diagrams and correlation

Concept

A **scatter diagram** plots paired values of two variables to show whether a relationship exists between them.

Correlation describes the strength and direction of a *linear* relationship between two variables.

Positive correlation means both variables tend to increase together. **Negative correlation** means one variable tends to increase as the other decreases. **No correlation** means there is no consistent linear pattern between them.

Correlation can be strong or weak, depending on how closely the points cluster around a straight line. However strong it is, correlation never on its own proves that one variable *causes* changes in the other.

Key vocabulary

Term	Meaning
Scatter diagram	A plot of paired (x, y) values for two variables
Positive correlation	As x increases, y tends to increase
Negative correlation	As x increases, y tends to decrease
No correlation	There is no consistent linear pattern between the variables
Causation	One variable directly causing a change in another — not implied by correlation alone

Procedure

1. Plot each pair of values as a point on the scatter diagram, with the explanatory (independent) variable on the horizontal axis.
2. Judge the direction of any pattern: do the points broadly rise, fall, or show no trend?
3. Judge the strength: do the points lie close to a straight line (strong) or are they widely scattered (weak)?

4. If asked to comment on causation, note explicitly that correlation alone does not establish it — a third, unmeasured variable could be responsible for both, or the relationship could be coincidental.

Worked example

EXAMPLE 4.1

A scatter diagram of ice cream sales against daily temperature shows the points lying close to a straight line that rises as temperature increases. (a) Describe the correlation shown. (b) A student claims this proves higher temperatures cause higher ice cream sales. Comment on this claim.

Step 1 (part a). The points show strong positive correlation between temperature and ice cream sales. **B1**

Step 2 (part b). Correlation does not imply causation — while a plausible causal link exists here (heat increases demand for cold food), the data alone does not prove it; other factors coinciding with hot days (e.g. more people outdoors) could also explain the pattern. **B1**

Answer (a) Strong positive correlation. (b) The scatter diagram shows an association, not proof of a causal link; other explanations remain possible.

EXAM TIP

It is easy to confuse “no correlation” with “no relationship at all” — two variables can have a strong non-linear relationship (e.g. a U-shape) while showing almost zero linear correlation. Remember that correlation coefficients specifically measure *linear* association.

4.2 The product moment correlation coefficient (PMCC)

Concept

The **product moment correlation coefficient**, r , gives a precise numerical measure of linear correlation, always between -1 and $+1$. A value of $+1$ means perfect positive linear correlation, -1 means perfect negative linear correlation, and 0 means no linear correlation. The closer $|r|$ is to 1 , the stronger the linear relationship. In exams, r is almost always calculated using a calculator's statistics mode from raw data, or found by substituting given summary statistics (S_{xx} , S_{yy} , S_{xy}) into the formula below.

Key vocabulary

Term	Meaning
PMCC (r)	A number from -1 to $+1$ measuring the strength and direction of linear correlation
S_{xx} , S_{yy} , S_{xy}	Summary quantities calculated from the raw data (sums of squared deviations and cross-products), used to find r and the regression line

Key formulas

KEY FORMULAS

First calculate three summary quantities from the raw data:

$$S_{xy} = \sum xy - \frac{\sum x \sum y}{n}$$

$$S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n}, \quad S_{yy} = \sum y^2 - \frac{(\sum y)^2}{n}$$

Then substitute these into the formula for r :

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$$

Procedure

1. Find (or use given values of) $\sum x$, $\sum y$, $\sum x^2$, $\sum y^2$, $\sum xy$, and n .
2. Calculate S_{xx} , S_{yy} and S_{xy} using the formulas above.
3. Substitute into the formula for r .
4. Interpret r in context: state the strength (weak/moderate/strong) and direction (positive/negative), referring to the actual variables in the question.

Worked example

EXAMPLE 4.2

For 8 paired observations: $\sum x = 40$, $\sum y = 96$, $\sum x^2 = 220$, $\sum y^2 = 1200$, $\sum xy = 500$, $n = 8$. Calculate the PMCC and interpret it.

Step 1. Calculate S_{xy} , S_{xx} and S_{yy} using the formulas above. **M1**

$$S_{xy} = 500 - \frac{40 \times 96}{8} = 20$$

$$S_{xx} = 220 - \frac{40^2}{8} = 20$$

$$S_{yy} = 1200 - \frac{96^2}{8} = 48$$

A1 A1 A1

Step 2. Substitute into the PMCC formula. **M1**

$$r = \frac{20}{\sqrt{20 \times 48}} = \frac{20}{\sqrt{960}} = 0.645$$

A1

Answer $r \approx 0.645$, indicating a moderate positive linear correlation between the two variables.

EXAM TIP

A very common mark scheme point: r must be interpreted *in context* with a strength descriptor (e.g. “moderate”, “strong”), not simply quoted as a bare number. Also flag that a strong r does not by itself justify extrapolating a regression line far beyond the range of the data — that’s a separate error, covered in 4.3.

4.3 Linear regression

Concept

While correlation measures how strong a linear relationship is, **regression** finds the actual equation of the best-fit line, allowing predictions of one variable from the other. The **explanatory (independent) variable**, x , is used to predict the **response (dependent) variable**, y . The regression line always passes through the mean point $(\bar{x}, \bar{y})$, and its gradient and intercept are found from the same S_{xx} and S_{xy} used for the PMCC.

Key vocabulary

Term	Meaning
Explanatory (independent) variable	The variable used to make predictions, plotted on the x-axis
Response (dependent) variable	The variable being predicted, plotted on the y-axis
Mean point	The point through which every regression line passes, found from the mean of the x-values and the mean of the y-values
Interpolation	Predicting a y-value for an x within the range of the original data (reliable)
Extrapolation	Predicting a y-value for an x outside the range of the original data (unreliable)

Key formulas

KEY FORMULAS

The regression line of y on x has the equation:

$$y = a + bx$$

where the gradient b and intercept a are found from:

$$b = \frac{S_{xy}}{S_{xx}}, \quad a = \bar{y} - b\bar{x}$$

Procedure

1. Calculate S_{xy} and S_{xx} (as in 4.2).
2. Find the gradient $b = S_{xy} \div S_{xx}$.
3. Find $\bar{x} = \Sigma x/n$ and $\bar{y} = \Sigma y/n$ (the mean point through which the line always passes), then find the intercept $a = \bar{y} - b\bar{x}$.
4. State the regression equation using the actual variable names from the context, not just x and y .
5. When using the line to predict, check whether the x -value is within the range of the original data (interpolation, reliable) or outside it (extrapolation, unreliable) — flag any extrapolation explicitly.

Worked example

EXAMPLE 4.3

Using the summary statistics from Example 4.2 ($\Sigma x = 40$, $\Sigma y = 96$, $n = 8$, $S_{xy} = 20$, $S_{xx} = 20$), find the equation of the regression line of y on x , and use it to estimate y when $x = 6$ (the original x -values ranged from 2 to 9).

Step 1. Find the gradient. **M1**

$$b = \frac{20}{20} = 1$$

A1

Step 2. Find the mean point and the intercept: $\bar{x} = 40/8 = 5$, $\bar{y} = 96/8 = 12$. **M1**

$$a = 12 - 1(5) = 7$$

A1

Step 3. State the equation and substitute $x = 6$.

$$y = 7 + x$$

$$y = 7 + 6 = 13$$

M1 A1

Answer Regression equation: $y = 7 + x$. Since $x = 6$ lies within the original data range (2 to 9), this is interpolation and the estimate $y = 13$ is considered reliable.

EXAM TIP

Always check the x -value against the range of the original data before quoting a prediction as reliable — this is a near-guaranteed mark on any regression question that asks for an estimate, and is one of the easiest to lose by skipping straight to the arithmetic.

4.4 Non-linear regression: reducing to a linear form

Concept

Many real relationships are not linear — for example exponential growth ($y = ab^x$) or a power law ($y = ax^n$). These can still be analysed using linear regression by taking logarithms of both sides, which turns the equation into the straight-line form $Y = A + BX$, where X and Y are new variables built from x and y — exactly which ones depends on the model, as shown below.

A straight-line fit can then be found for the transformed data, and the original constants recovered by reversing the logarithm at the end.

Key vocabulary

Term	Meaning
Reducing to linear form	Transforming a non-linear model into a straight-line equation using logarithms
Power law	A model of the form $y = ax^n$, where a and n are constants

Term	Meaning
Exponential model	A model of the form $y = ab^x$, where a and b are constants

Key formulas

KEY FORMULAS

For $y = ax^n$, taking logs of both sides:

$$\log y = \log a + n \log x$$

which is linear in $\log x$ and $\log y$, with gradient n and intercept $\log a$.

For $y = ab^x$, taking logs of both sides:

$$\log y = \log a + x \log b$$

which is linear in x and $\log y$, with gradient $\log b$ and intercept $\log a$.

Procedure

1. Identify which non-linear model is being tested (power law $y = ax^n$, or exponential $y = ab^x$) — the question usually states this.
2. Take logarithms of both sides and rearrange to match the linear form $Y = A + BX$ above, deciding which variables need transforming.
3. Use the transformed data (or a given regression line of the transformed data) to find the gradient and intercept, exactly as in 4.3.
4. For a power-law model ($y = ax^n$): the gradient equals n directly. Recover a from the intercept using $a = 10^{(\text{intercept})}$ (or $e^{(\text{intercept})}$ if natural logarithms were used).
5. For an exponential model ($y = ab^x$): recover b from the gradient using $b = 10^{(\text{gradient})}$. Recover a from the intercept using $a = 10^{(\text{intercept})}$ (again matching whichever log base was used).

Worked example

EXAMPLE 4.4

A scientist believes two variables are related by $y = ax^n$. Taking logs (base 10) of the data gives a regression line of $\log y$ on $\log x$: $\log y = 0.30 + 2 \log x$. Find the values of a and n .

Step 1. Compare to $\log y = \log a + n \log x$: the gradient is n and the intercept is $\log a$. **M1**

$$n = 2$$

A1

Step 2. Convert the intercept back to a .

$$\log a = 0.30 \Rightarrow a = 10^{0.30} = 2.00 \quad (3 \text{ s.f.})$$

M1 A1

Answer $n = 2$ and $a \approx 2.00$, so $y \approx 2.00x^2$.

EXAM TIP

It is common to correctly find the gradient and intercept of the transformed line but then forget to convert the intercept back using a power of 10 (or e , depending on the log base used) to recover a — the gradient usually needs no further conversion for a power-law model, but the intercept always does. Get into the habit of writing out “log a = intercept” as an explicit line before solving for a .

Chapter summary

Idea	Key point
Correlation	Direction and strength of a <i>linear</i> relationship; never proves causation
PMCC (r)	$r = S_{xy} / \sqrt{(S_{xx} \times S_{yy})}$; between -1 and $+1$, interpreted in context
Regression line	$y = a + bx$, with $b = S_{xy}/S_{xx}$ and $a = \bar{y} - b\bar{x}$
Interpolation vs extrapolation	Predicting within the data range is reliable; outside it is not
Reducing to linear form	Take logs of $y = ax^n$ or $y = ab^x$ to get a straight-line equation, then convert back

Common exam mistakes

- Quoting the PMCC as a bare number without a strength/direction interpretation in context.
- Claiming correlation proves causation.
- Using a regression equation to extrapolate far beyond the data range without flagging the estimate as unreliable.
- Forgetting to convert the transformed intercept back (via a power of 10 or e) when recovering the original constant in a reduced-to-linear-form model.

You should now be able to

- Describe the correlation shown by a scatter diagram and discuss whether it implies causation.
- Calculate the PMCC from raw data or given summary statistics, and interpret it in context.
- Find the equation of a regression line and use it to make predictions, judging reliability by interpolation vs extrapolation.
- Reduce a power-law or exponential relationship to a linear form using logarithms, and recover the original constants.